higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

LOGIC SYSTEMS N5

6 AUGUST 2019

This marking guideline consists of 6 pages.

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QUESTION 1

			4 A	2 B	1 C	J_A	K_A	J_B	K_B	J_C	K_C	
→	→	↓	0	0	1	0	Ø	1	Ø	Ø	1	
↑		↓	0	1	0	0	Ø	Ø	0	1	Ø	
↑		↓	0	1	1	1	Ø	Ø	1	Ø	1	Λ
↑		↓	1	0	0	Ø	0	0	Ø	1	Ø	
↑		↓	1	0	1	Ø	0	1	Ø	Ø	1	
↑	←	←	1	1	0	Ø	1	Ø	1	1	Ø	
↑	←	←	0	0	0	0	Ø	0	Ø	1	Ø	Λ
↑	←	←	1	1	1	Ø	1	Ø	1	Ø	0	

$Q_n \rightarrow Q_n + 1$	J	K
0 → 0	0	Ø
0 → 1	1	Ø
1 → 0	Ø	1
1 → 1	Ø	0

J_A

/c \ /A/B	00	01	11	10
0	0	0	Ø	Ø
1	0	1	Ø	Ø

$J_A = BC$ Λ

K_A

/c \ /A/B	00	01	11	10
0	Ø	Ø	1	0
1	Ø	Ø	1	0

$K_A = B$ Λ

J_B

/c \ /A/B	00	01	11	10
0	0	Ø	Ø	0
1	1	Ø	Ø	1

$J_B = C$ Λ

K_B

/c \ /A/B	00	01	11	10
0	Ø	0	1	Ø
1	Ø	1	1	Ø

$Kb = A + C$ Λ

J_C

/c \ /A/B	00	01	11	10
0	1	1	1	1
1	Ø	Ø	Ø	Ø

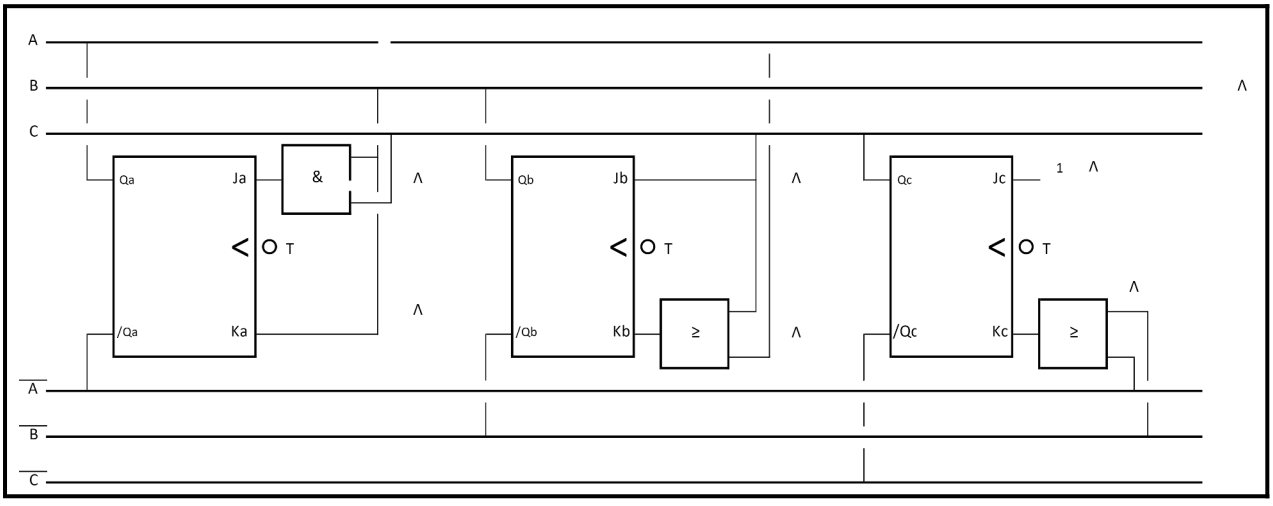
$J_C = 1$ Λ

K_C

/c \ /A/B	00	01	11	10
0	Ø	Ø	Ø	Ø
1	1	1	0	1

$K_C = /A + /B$ Λ

(7)

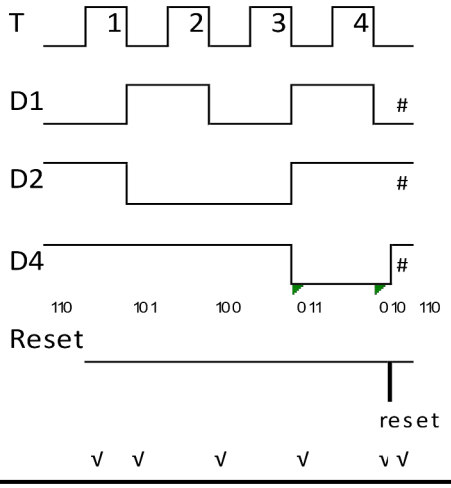


(8)

[15]

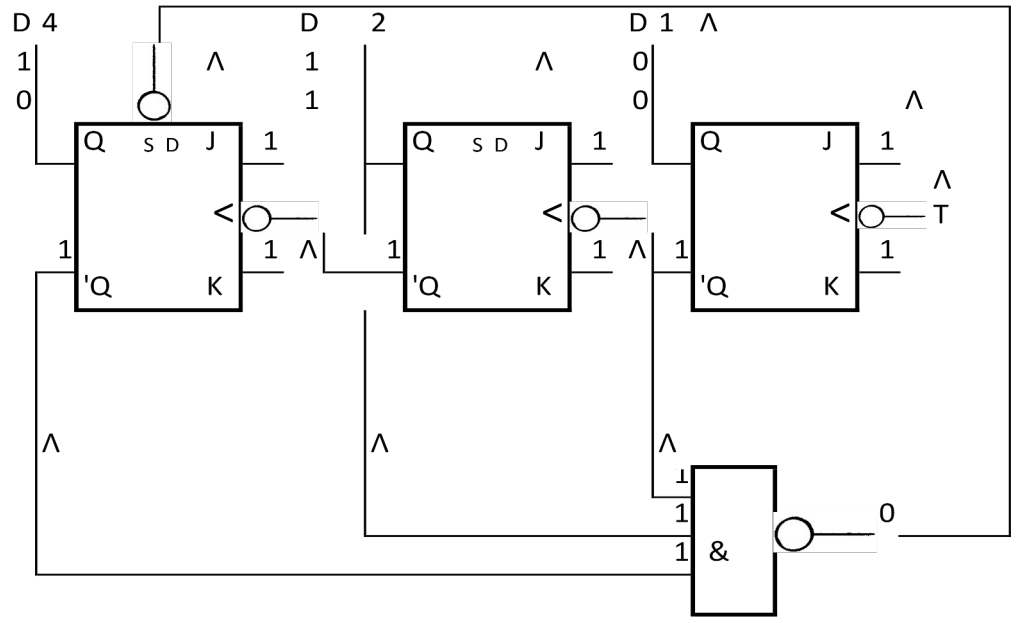
QUESTION 2

D4	D2	D1	T	
1	1	0	0	√
1	0	1	1	√
1	0	0	2	
0	1	1	3	√
0	1	0	4	
1	1	0	4	√



√ √ √ √ √ √

(10)



(10 × ½) (5)

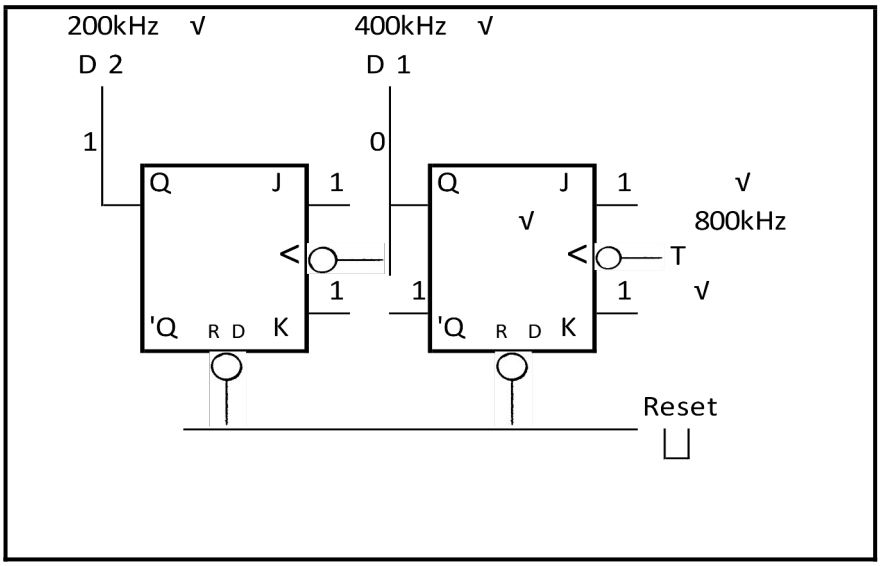
[15]

QUESTION 3

3.1
- Time measurement
- Totalising
- Frequency
- Period measurement
- Frequency ratio

3.2 It generates the gate open signal by dividing the reference oscillator frequency by a factor of 10.

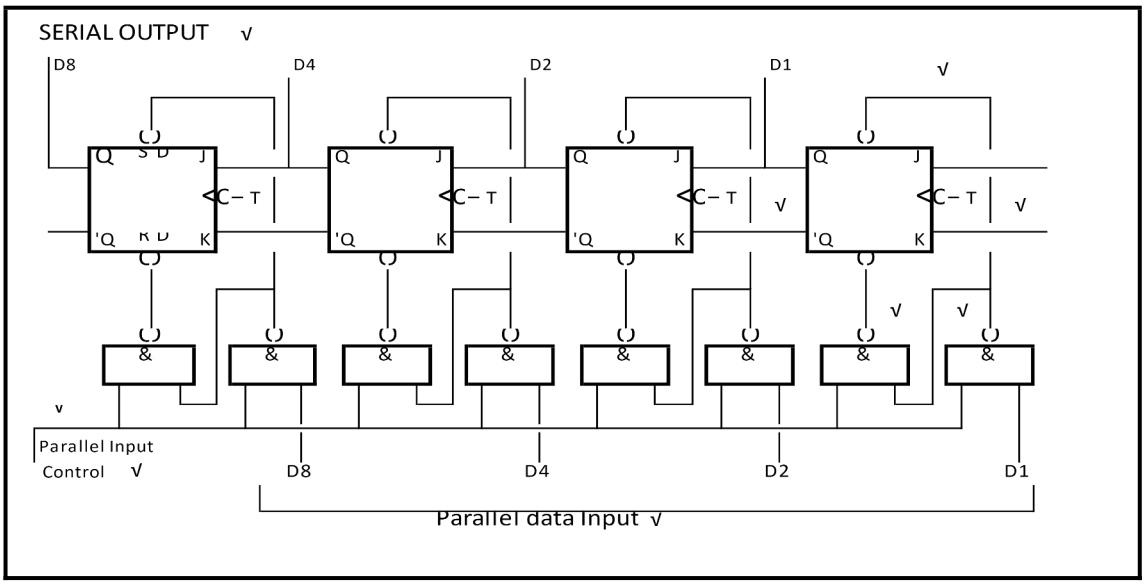
3.3



(3 × 5) **[15]**

QUESTION 4

4.1



(9)

4.2 Synchronous (1)

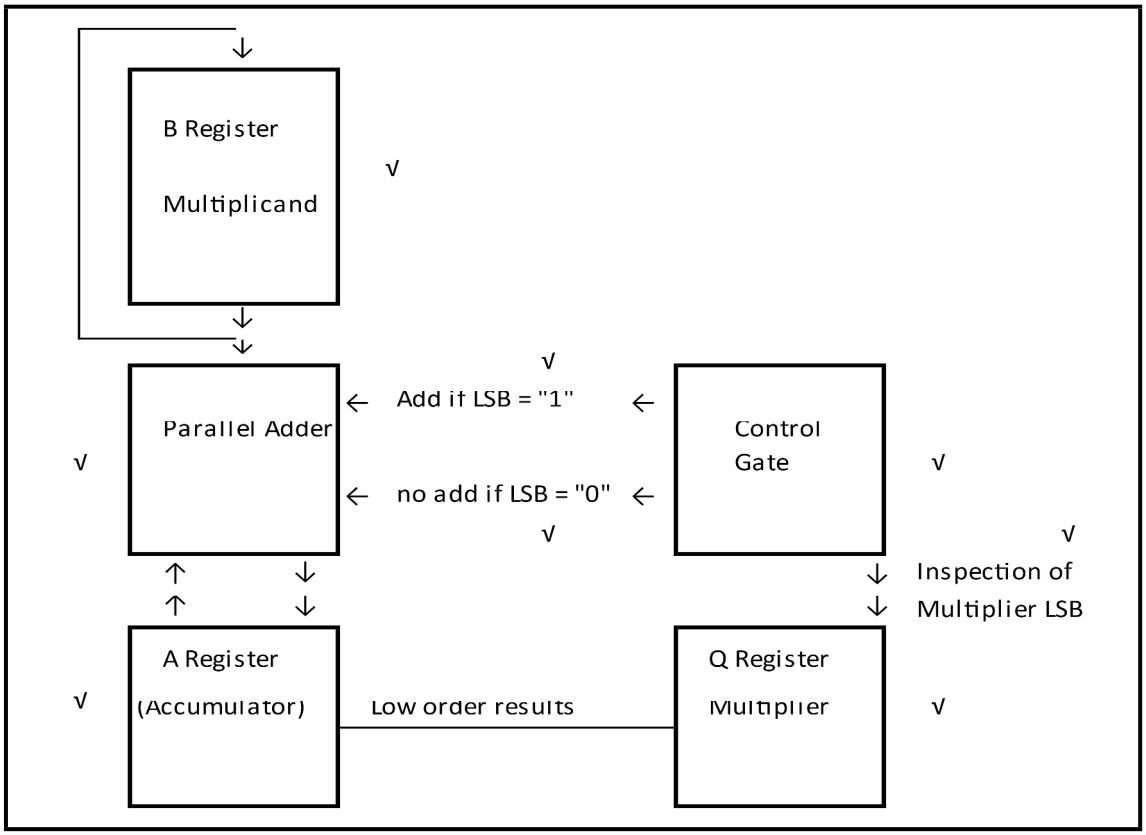
4.3 One (1)

4.4 Four (1)

[12]

QUESTION 5

5.1



(8)

5.2

- If the LSB of the multiplier is 1, the multiplicand is added to the accumulator register contents.
- If the LSB of the multiplier is 0, the multiplicand is not added to the accumulator register contents.
- After the add or no-add, the accumulator and Q register contents are shifted right.
- The above cycle is repeated for the rest of the bits in the multiplier.
- The final answer is in the accumulator register. (6)

[14]

QUESTION 6

6.1
- The computer stores the program instructions
- and data in the SRAM or DRAM
- when the computer is running and
- when the computer is not running
- the program instructions and data are stored in the hard drive. (5)

6.2

6.2.1	E
6.2.2	G
6.2.3	J
6.2.4	I
6.2.5	C
6.2.6	A
6.2.7	D
6.2.8	B
6.2.9	F
6.2.10	H

(10 × 1) (10)
[15]

QUESTION 7

7.1
- Laser
- Inkjet
- Graph (3)

7.3
- Laser
- Inkjet
- Graph
- Thermal
- Static (5)

7.2
- Drum
- Chain
- Dot-matrix (3)

7.4 Laser (1)

7.5
- Thermal
- Dot-matrix (2)

[14]

TOTAL: 100

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